

Technical Design Document

Programming_Excercise_1

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Program File:	programming_excercise_1.py
Program Description	This program simulates a cinema ticket presale. It starts with 20 available tickets, allows each buyer to purchase between 1 and 4 tickets, validates each purchase, updates the number of tickets remaining, and counts the total number of successful buyers until all tickets are sold.

#Program Constants

TOTAL_TICKETS (int) : The starting number of tickets available for the presale. The value is 20.

MAX_TICKETS_PER_BUYER (int) : The maximum number of tickets one buyer may purchase at a time. The value is 4.

#Function: get_ticket_request

Description: Prompts the user to enter how many tickets they would like to buy and returns the number entered.

Parameters:

- **None:** This function does not receive any parameters.

Variables:

- **tickets_requested (int):** Stores the number of tickets the user wants to buy.

Logic:

1. Display a prompt asking how many tickets the user would like to buy.
2. Convert the user input to an integer.
3. Store the value in tickets_requested.
4. Return tickets_requested to the calling function.

Return: int: The number of tickets requested by the buyer.

#Function: process_purchase

Description: Subtracts the number of tickets requested from the current number of tickets remaining.

Parameters:

- **tickets_remaining (int):** The number of tickets available before the current purchase.
- **tickets_requested (int):** The number of tickets the current buyer wants to purchase.

Variables:

- **tickets_remaining (int):** Updated number of tickets after the purchase is processed.

Logic:

5. Subtract tickets_requested from tickets_remaining.
6. Store the updated value in tickets_remaining.
7. Return the updated tickets_remaining value.

Return: int: The updated number of tickets remaining after the purchase.

#Function: main

Description: Controls the overall ticket presale process. It initializes the ticket count and buyer count, repeatedly asks users for purchases until all tickets are sold, validates each purchase, and displays the final number of buyers.

Parameters:

- **None:** This function does not receive any parameters.

Variables:

- **tickets_remaining (int):** Tracks how many tickets are still available.
- **total_buyers (int):** Counts how many successful buyers purchased tickets.
- **tickets_requested (int):** Stores the number of tickets requested by the current buyer.

Logic:

8. Set tickets_remaining equal to TOTAL_TICKETS.
9. Set total_buyers to 0.
10. While tickets_remaining is greater than 0, call get_ticket_request to get the buyer request.
11. Check that the request is at least 1 ticket, no more than MAX_TICKETS_PER_BUYER, and no more than the tickets remaining.
12. If the request is valid, call process_purchase, update tickets_remaining, add 1 to total_buyers, and display the tickets remaining.
13. If the request is invalid, display an error message explaining the buying rules.
When tickets_remaining reaches 0, display the sold-out message and total number of buyers.

Return: None

#Input Validation

- The user must request at least 1 ticket.
- The user may not request more than 4 tickets at one time.
- The user may not request more tickets than the amount remaining.
- Invalid requests do not reduce the ticket count and do not increase the buyer count.

#Program Flow Summary

Start with 20 tickets.

Ask each buyer how many tickets they want.

Validate the requested amount.

Subtract valid purchases from the total tickets remaining.

Repeat until the number of tickets remaining is 0.

Print the total number of successful buyers.

#Repository Link

Link to repository: [Respository Link](#)

#Output Screenshot



```
Run programming_exercise_1 x
"C:\Users\The Computer Place\PyCharmMiscProject\.venv\Scripts\python.exe" "C:\Users\The Computer Place\PyCharmMiscProject\programming_exercise_1.py"
How many tickets would you like to buy?4
Tickets remaining: 16
How many tickets would you like to buy?6
Invalid purchase. You may buy 1 to 4 tickets, and no more than the remaining amount.
How many tickets would you like to buy?3
Tickets remaining: 13
How many tickets would you like to buy?3
Tickets remaining: 10
How many tickets would you like to buy?3
Tickets remaining: 7
How many tickets would you like to buy?3
Tickets remaining: 4
How many tickets would you like to buy?4
Tickets remaining: 0
Hey! Wait a minute...You bought all the tickets for yourself! Greedy! .
Total buyers: 6

Process finished with exit code 0
```